
Mixing Matters: Evidence Position Bias Across Attention and State-Space Language Models

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Abstract

1 Language models often answer multi-document questions less accurately when
2 the relevant document is placed in the middle rather than near the beginning or
3 end. We test whether this pattern differs between models that use attention and
4 models from the Mamba family, which process context through a running state.
5 For each question, we move the answer-bearing document through ten positions
6 while keeping the question and nine distracting documents fixed. Our main com-
7 parisons use 800 exploratory questions; smaller task and prompt checks use 50 or
8 200 examples, and the larger held-out set has not been examined. Pythia-2.8B was
9 about five percentage points more accurate when the answer appeared near the be-
10 ginning than in the middle, whereas two similarly sized, Pile-trained Mamba mod-
11 els showed no such beginning advantage. A more closely matched 8B comparison
12 pointed in the same direction: a hybrid Mamba-2 model with a small number of at-
13 tention layers showed a larger beginning advantage than the pure Mamba-2 model,
14 but the difference between them remains too uncertain for a firm conclusion. The
15 contrast appears in larger but not the smallest model pairs, is not explained by the
16 tested change of training corpus, and reappears for Pythia on a synthetic retrieval
17 task. Prompt changes can strongly alter the pattern, and exploratory internal-state
18 analyses show that Mamba retains position information even without the begin-
19 ning advantage. These findings suggest that model design affects how available
20 evidence is used across a context, but they do not show that attention itself is the
21 cause.

22 1 Introduction

23 Transformers often read multi-document contexts unevenly: accuracy peaks when the answer-
24 bearing document is near an edge and falls in the middle (Liu et al., 2024). This behavior matters
25 because retrieval systems cannot assume that all evidence positions are equally usable. Long-context
26 benchmarks expose related failures across question answering, summarization, and retrieval (Bai et
27 al., 2024; Shaham et al., 2023; Li et al., 2024). Prior work links lost-in-the-middle behavior to
28 U-shaped attention patterns (Hsieh et al., 2024a), but model-family comparisons still confound se-
29 quence mixing with data, tokenizer, scale, depth, and positional encoding (Su et al., 2024; Press et
30 al., 2022). State-space models such as Mamba (Gu and Dao, 2024; Dao and Gu, 2024) provide a
31 useful contrast because they compress the prefix into a recurrent state instead of using dense causal
32 attention.

33 We make three contributions. First, we build an auditable ten-position evaluation with floor and
34 ceiling anchors, preserved generations, and inference settings recorded per example. Second, we
35 find a paired architecture-position interaction in a Pile-trained 2.8B comparison and test it with a
36 more tightly matched 8B pure/hybrid pair. Third, we organize scale, corpus, task, prompt, sink,

and probe results into mechanistic evidence with explicit uncertainty. All reported results use an exploratory split; the held-out confirmatory split remains unspent.

2 Related work

Position bias in long contexts Liu et al. (2024) established the lost-in-the-middle curve in retrieval and multi-document QA. This task derives its questions from Natural Questions (Kwiatkowski et al., 2019), while LongBench, ZeroSCROLLS, and LooGLE broaden long-context evaluation across tasks, languages, and reasoning demands (Bai et al., 2024; Shaham et al., 2023; Li et al., 2024). Subsequent work measures effective context length (Hsieh et al., 2024b), associates the curve with attention allocation (Hsieh et al., 2024a), and proposes reordering, compression, retrieval, and context-use interventions (An et al., 2024; Peysakhovich and Lerer, 2023; Jiang et al., 2024; Xu et al., 2024; Agrawal et al., 2024). Irrelevant context can itself distract a model, motivating our use of the same distractors at every tested position (Shi et al., 2023). We focus on whether the curve changes across sequence mixers.

Sequence mixers Dense self-attention (Vaswani et al., 2017) mixes token pairs under a causal mask and commonly encodes order with rotary embeddings or linear attention biases (Su et al., 2024; Press et al., 2022). Alternative long-sequence mixers include structured state spaces, recurrent weighted key-value models, and long convolutions (Gu et al., 2022; Peng et al., 2023; Poli et al., 2023). Mamba and Mamba-2 carry information through selective state-space recurrences (Gu and Dao, 2024; Dao and Gu, 2024). Hybrid models such as Griffin, Jamba, and Nemotron-H combine recurrent or state-space blocks with attention (De et al., 2024; Lieber et al., 2024; NVIDIA, 2025). Waleffe et al. (2024) release the pure and hybrid 8B Mamba-2 checkpoints used in our tightest controlled comparison.

Attention sinks and probes Xiao et al. (2024) showed that causal transformers can concentrate attention on initial tokens, and Sun et al. (2024) connected unusually large internal activations to a small set of tokens. We test whether this attention-sink statistic covaries with primacy, treating the result as correlational because a valid sink-blocking ablation is absent. Our position probe follows the principle that a diagnostic classifier needs a control task before its accuracy can be interpreted as evidence about a representation (Hewitt and Liang, 2019).

3 Measurement and statistics

Task and anchors We use the 2,655 released questions from the Lost-in-the-Middle 10-document QA setting (Liu et al., 2024), which is built from Natural Questions (Kwiatkowski et al., 2019). Each question has one gold document and nine distractors. We build ten prompts per question, moving only the gold document through positions 0 to 9 while retaining the same distractors and token budget. A no-gold floor measures guessing or memorized knowledge, and a gold-only ceiling measures answerability under perfect retrieval.

Edges and statistics Let a_S , a_M , and a_E be mean accuracy at start positions $\{0, 1\}$, middle positions $\{4, 5\}$, and end positions $\{8, 9\}$. We define primacy as $a_S - a_M$ and recency as $a_E - a_M$. The question is the unit of analysis. Each of 10,000 bootstrap samples draws complete ten-position question bundles with replacement, following the bootstrap principle of resampling the independent unit (Efron, 1979), and model comparisons reuse the same sampled questions. We report percentile 95% confidence intervals and apply Holm’s sequential correction to the primacy and recency tests within each comparison family (Holm, 1979). The fixed split contains 800 exploratory and 1,855 confirmatory questions. Only the exploratory split is reported.

Reproducibility and controls The harness vendors the official prompt builder and best_subspan_em scorer at a pinned commit. It also records normalized exact match, following the normalization convention popularized by SQuAD (Rajpurkar et al., 2016), and first-line extraction. Generation is greedy with temperature 0, at most 32 new tokens, fixed seeds, BF16 where supported, pinned model revisions, and one Mamba execution path. Each unique run preserves prompts, raw generations, failures, token counts, software versions, and decoding settings. The released key-value

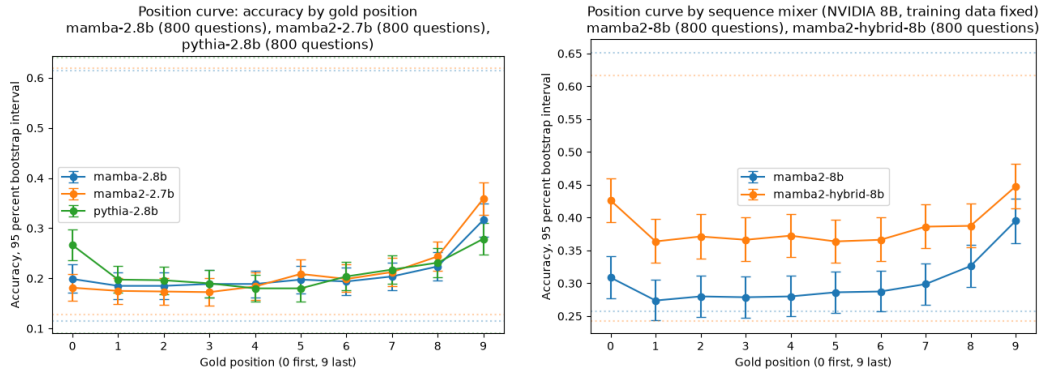
task provides an executed positive control: the pipeline detects a +0.38 Pythia start-versus-middle effect. On 200 exploratory Pythia questions, sham-gold accuracy was 0.1235, within 0.0285 of the 0.0950 closed-book floor; primacy was +0.0050 (95% CI $[-0.0325, 0.0425]$) and recency was -0.0175 (95% CI $[-0.0500, 0.0125]$). The distractor-order control’s permutation accuracies were 0.2850, 0.3000, and 0.2767, a maximum spread of 0.0233. Both controls passed their prespecified gates on Pythia; they do not test the 8B Megatron checkpoints.

4 Architecture comparisons

Primary 2.8B comparison We compare pythia-2.8b (Biderman et al., 2023) with mamba-2.8b and mamba2-2.7b. All are base models pretrained on the Pile (Gao et al., 2020), have similar parameter counts, and use related tokenizers. Table 1 and Figure 1 show a positive primacy edge for Pythia and no positive primacy edge for either Mamba model. The paired Mamba-minus-Pythia primacy interactions are -0.053 and -0.070 (both Holm $p < 10^{-4}$), while Mamba-1 versus Mamba-2 is uncertain ($+0.017$, 95% CI $[-0.005, 0.038]$). All three models have positive recency edges. These comparisons match corpus and approximate scale, but not depth, positional encoding, state size, or checkpoint family.

Table 1: Pile-trained models on 800 exploratory questions. Intervals use a question-level paired bootstrap; p is Holm-corrected.

Model	Primacy (95% CI)	Recency (95% CI)	Floor / Ceiling
pythia-2.8b	+0.052 [0.032, 0.072], $p < 10^{-4}$	+0.075 [0.054, 0.097], $p < 10^{-4}$	0.091/0.640
mamba-2.8b	-0.001 $[-0.016, 0.014]$, $p=0.91$	+0.077 [0.058, 0.097], $p < 10^{-4}$	0.115/0.615
mamba2-2.7b	-0.018 $[-0.034, -0.003]$, $p=0.02$	+0.105 [0.086, 0.124], $p < 10^{-4}$	0.128/0.619



(a) Pile-trained models near 2.8B parameters

(b) Matched pure and hybrid Mamba-2 models at 8B

Figure 1: Accuracy as the answer-bearing document moves through the context on 800 exploratory questions. Pythia rises at the beginning, while both smaller Mamba variants do not; the matched 8B hybrid also rises more than the pure model, but their paired difference remains uncertain. Dotted lines show each model’s no-gold floor and gold-only ceiling.

Matched 8B attention comparison Our strongest control uses NVIDIA’s pure and hybrid 8B Mamba-2 checkpoints (Waleffe et al., 2024), which share training data, tokenizer, scale, depth, and positional setup. The hybrid contains 7% attention layers. Within models, the pure model’s primacy edge is uncertain ($+0.008$, $p = 0.36$) and the hybrid’s is positive ($+0.027$, $p = 0.007$). That difference of significance is not itself a significant difference. The paired hybrid-minus-pure primacy effect is $+0.019$ (95% CI $[-0.006, 0.044]$, Holm $p = 0.144$), while the recency effect is -0.028 (95% CI $[-0.055, -0.001]$, Holm $p = 0.089$). This direction supports the 2.8B observation but does not isolate a causal attention effect.

Table 2: Matched 8B primacy comparison on 800 exploratory questions. The paired interaction is the relevant architecture test.

Contrast	Primacy	95% CI	Holm p
pure Mamba-2	+0.008	$[-0.009, 0.024]$	0.362
hybrid	+0.027	$[0.008, 0.047]$	0.007
hybrid minus pure	+0.019	$[-0.006, 0.044]$	0.144

109 5 Scope checks

110 **Scale** Across five approximate size pairs, the Pythia-minus-Mamba primacy gap is near zero at
 111 the two smallest scales and is +0.062, +0.069, and +0.053 in the three larger pairs (Figure 2).
 112 The smallest Pythia ceiling is only 0.009, so this is a descriptive capability-and-family trend, not
 113 evidence for a depth threshold.

114 **Training corpus** Holding the Mamba-2.8B architecture fixed, the Pile-minus-SlimPajama primacy
 115 effect is -0.007 (95% CI $[-0.030, 0.016]$, $p = 0.568$) and the recency effect is -0.016 (95% CI
 116 $[-0.044, 0.012]$, Holm $p = 0.502$) (Soboleva et al., 2023). The floor differs substantially (0.270
 117 versus 0.114), so this comparison finds no edge-shape change but does not rule out training-data
 118 effects (Figure 2).

119 **Synthetic retrieval** On 50 RULER niah_single_1 examples at 2K tokens (Hsieh et al., 2024b),
 120 Pythia again has a primacy edge (+0.12, 95% CI $[0.05, 0.20]$). Both Mamba models score 1.0 at
 121 every depth, so this check reproduces the transformer pattern but cannot compare mixer curves on
 122 the synthetic task (Figure 3).

123 **Descriptive system comparison** We also run Llama-3.1-8B (Dubey et al., 2024), Qwen2.5-7B
 124 (Yang et al., 2024), and Nemotron-H-8B (NVIDIA, 2025). All three show positive primacy edges
 125 (+0.076, +0.041, +0.067). Only Llama versus Qwen has a significant paired primacy interaction
 126 (+0.036, Holm $p = 0.016$). Because the systems differ on architecture, data, tokenizer, token count,
 127 depth, and positional encoding, these results establish prevalence, not an architectural explanation
 128 (Figure 4).

129 6 Mechanistic evidence and intervention

130 **Attention sinks track primacy** Across the Pythia family, the 160M model has almost no token-
 131 0 attention sink (final-layer share 0.003), while the 2.8B model reaches 0.455 (Xiao et al., 2024).
 132 Pythia-410M has a small final-layer sink (0.034) and no significant primacy edge. Figure 3 is con-
 133 sistent with a sink mechanism, but the measurements share model scale as a confound and no valid
 134 sink ablation was completed.

135 **Utilization, not storage, is a hypothesis** A balanced linear probe on frozen mid-depth states pre-
 136 dicted edge versus middle position above its shuffled-label control for Pythia (0.603 versus 0.484)
 137 and Mamba (0.648 versus 0.508). Thus position information remains linearly decodable even when
 138 Mamba has no positive primacy edge. This supports a utilization hypothesis, but probe decodability
 139 does not prove that the model stores or uses the answer-bearing content.

140 **Training-free prompt intervention** On 200 exploratory questions, repeating the question both be-
 141 fore and after the documents increases Mamba primacy from 0.000 to +0.103 ($p < 10^{-4}$). Placing
 142 the question only before the documents does not create primacy (+0.020, $p = 0.366$), but reduces re-
 143 cency from +0.070 to +0.010. The intervention is training-free and practically useful, but it changes
 144 query repetition and token placement together, so it is not a clean test of fixed-state compression.

145 **Prompt sensitivity** Changing the prompt template reduces Pythia primacy from +0.070 to +0.005
 146 in the 200-question mechanism subset. This result warns against treating the measured magnitude
 147 as a template-invariant architectural constant.

148 7 Discussion and limitations

149 The most secure result is the 2.8B exploratory architecture-position interaction. The matched 8B
 150 direction, sink correlation, and probe results jointly make attention-mediated utilization a plausible
 151 hypothesis, while prompt sensitivity shows that architecture is not the only determinant.

152 Several limitations set the confirmatory boundary. All analyses use the exploratory split, so the
 153 planned 1,855-question test must be run under a frozen analysis before confirmatory language is
 154 warranted. The 2.8B and scale comparisons retain depth, positional-encoding, state-size, and capa-
 155 bility confounds. The matched 8B paired contrast crosses zero after Holm correction, and its run
 156 manifest records a dirty source tree, weakening exact reproducibility. These certification controls
 157 passed on Pythia, but they do not test the 8B Megatron checkpoints. A 50-generation blinded audit
 158 sample exists, but human labels were not completed. Finally, the attempted hybrid sink block did
 159 not alter the model because its custom attention path did not honor the hook; it is not an ablation
 160 result. The next decisive study should freeze exclusions and contrasts, repair the hybrid intervention,
 161 and evaluate the held-out split once.

162 8 Conclusion

163 On 800 exploratory Lost-in-the-Middle questions, Pythia-2.8B and two Pile-trained Mamba models
 164 have different primacy profiles under a paired ten-position evaluation. A matched 8B comparison is
 165 directionally consistent but statistically uncertain, so the present evidence supports an architecture-
 166 position interaction without establishing that attention causes it. The held-out split and the remaining
 167 control work provide a clear path from this exploratory workshop result to a confirmatory study.

168 A Experiment artifacts

169 The repository preserves raw generations, environment records, machine-readable summaries, and
 170 plots for every reported phase. The panels below reproduce the exact committed artifact images
 171 without redrawing or smoothing them. They summarize scope and mechanism checks that support
 172 interpretation but do not upgrade the exploratory results to confirmatory or causal evidence.

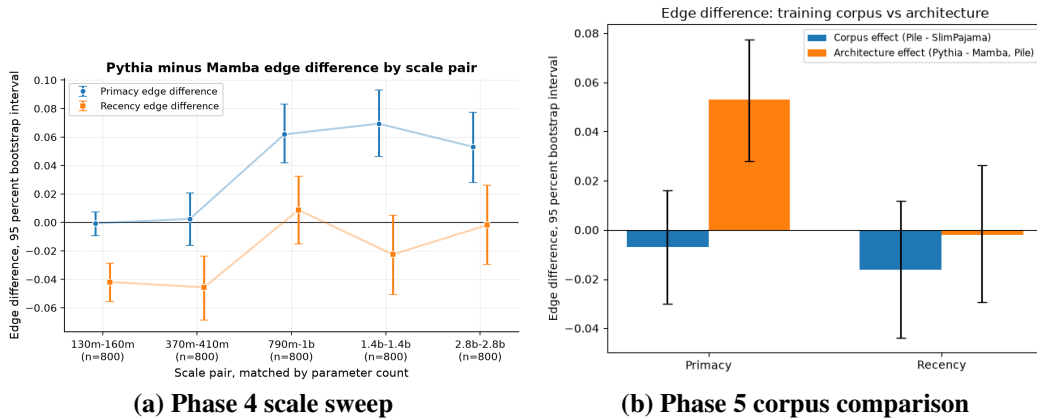


Figure 2: Exploratory scope checks. The scale sweep is descriptive because model capability and family differences vary together. The corpus comparison holds the Mamba architecture fixed and finds no clear change in curve shape, despite a substantial change in overall accuracy.

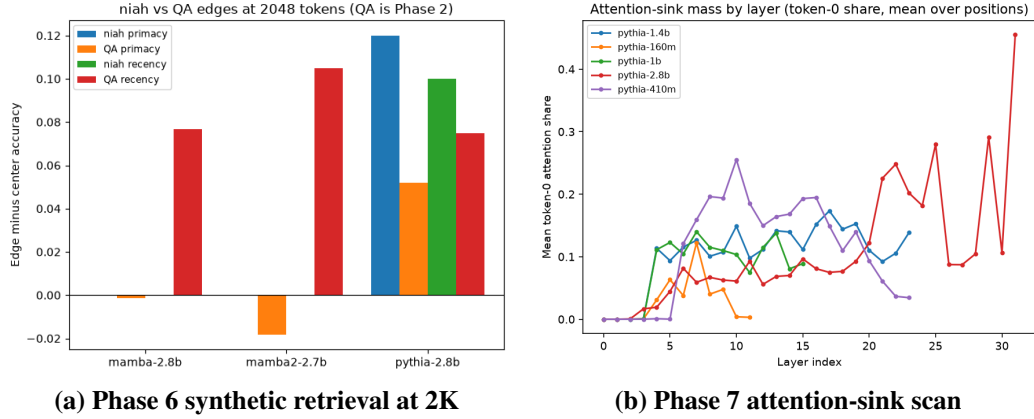


Figure 3: Exploratory task and mechanism artifacts. Pythia has a middle dip on the 2K synthetic task, while both Mamba models saturate and therefore cannot reveal a comparable curve. Late-layer token-0 attention grows with Pythia scale and covaries with the beginning advantage, but scale is a shared confound and no valid causal ablation was completed.

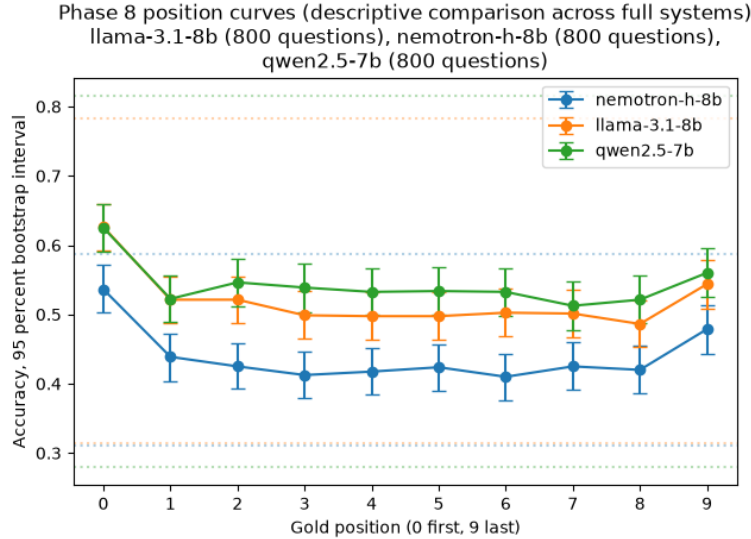


Figure 4: Phase 8 descriptive production-system comparison. All three systems show a beginning advantage, but differences in architecture, data, tokenizer, training, and capability prevent attribution to a single design choice.

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